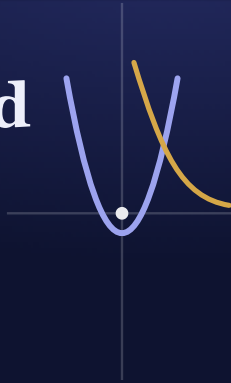


CLASS SYLLABUS · 2026-2027

Principles of Applied Engineering

The work you show on paper is how it sticks. Here's how our room runs and what I expect.

Mr. Gavaldon · Da Vinci School



DAILY How the studio runs

- 1 Grab your **assigned laptop**
- 2 **Sign in** to the program we're using
- 3 **Start work** — pick up where you left off
- 4 **Wait for instruction** when I call the class

PHONES

Up at the **start, every day**. Like the movies: phone away so everyone can focus. **I'll ask once** — then an email home, then a parent + admin meeting. Goal: **never** need it.

TUTORING & NOT FAILING

Fail a test → **one week of mandatory tutoring** (the department or me, on tutoring days). It's how we keep you from failing. Seats are assigned.

QUALITY OF WORK

- ✓ Finish your work **on your own**.
- ✓ Classwork done **in class**, the day it's assigned.
- ✓ On THIS class? No other classes or activities.
- ✓ **No AI in class or on tests** — you have to know it.

NOT ACCEPTED

- ✗ Yelling, fighting, hitting/touching others
- ✗ Eating, cussing, out of uniform
- ✗ Not turning in / not working in class
- ✗ No-showing tutoring; table-hopping; daily restroom

CORE 4

Respect

Responsibility

Quality of Work

Quality of Self

Effort you can see.

Course Roadmap

Principles of Applied Engineering
tentative – may shift

SEMESTER 1 · FALL

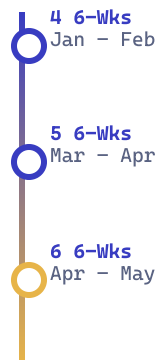


Foundations Bootcamp 10w
Foundations Bootcamp front-loaded TEKS; Sketching, Part Modeling

Fusion Advanced Modeling & Sculpt 5w
Sweep/Loft, Sculpt, Direct

Engineering Systems & Careers 3w
Engineering Systems & Careers robotics/automation + electrical & mechanical — EXPLICIT d 2...
★ Semester 1 Final

SEMESTER 2 · SPRING



Fusion Assemblies & Drawings 7w
Assembly, Drawing — reinforces systems d 2, d 7, d 8; ACU Certification Prep + Exam

Capstone — Open Design Challenge /... 6w
Capstone — Open Design Challenge / InSPIRESS competition project

Python — project-based 6w
Python — project-based coding track
★ Final Exam

BRING EVERY DAY

- ✓ A pencil or pen
- ✓ A pencil sharpener
- ✓ A glue stick
- ✓ A folder just for this class
- ✓ A composition notebook
- ✓ Colored pencils or markers
- ✓ Scissors
- ✓ A mouse

For the class — bring one: hand sanitizer, a box of Kleenex, or disinfecting wipes.

iPads are allowed for class work. The first time I see texting, it goes away — and it does not come back out.

The 4-Year Engineering Pathway

CTE Levels 1-4 · Grades 9-12 ·
leading to certification

1
Grade 9

Principles of Applied Engineering **INTRODUCTORY** \$127.781
Engineering design process, safety, and **Autodesk Fusion CAD from week 2** — project past project.
▲ Fusion part-modeling portfolio ♦ Starts ACU: Sketching · Part Modeling

2
Grade 10

Manufacturing Engineering **INTERMEDIATE** \$127.813
Design → make: 3D printing/CNC, quality, and mechanical/electrical & automation systems.
▲ Prototype-to-production build ♦ Adds Assembly · Drawings · OSHA-10 (opt.)

3
Grade 11

Engineering Design & Problem Solving · InSPIRESS 1 **ADVANCED** \$127.785
Full design process on a real problem; **design a payload and compete** (InSPIRESS / UAH).
▲ InSPIRESS competition entry ♦ Sit the ACU — Fusion exam

4
Grade 12

Engineering Design & Presentation II · InSPIRESS 2 **CAPSTONE** \$127.784
Capstone: advanced system design + presentation; lead a competition-grade design cycle; industry-ready senior portfolio.
▲ InSPIRESS capstone + portfolio ♦ ACU earned · stretch ACP / dual credit

CERTIFICATION Autodesk Certified User — Fusion

Certiport · 35-40 Q · 50 min · pass 700/1000

Levels follow the Texas CTE Program of Study (Level 1 intro → Level 4 capstone). Course codes tentative — confirm with the campus CTE coordinator.